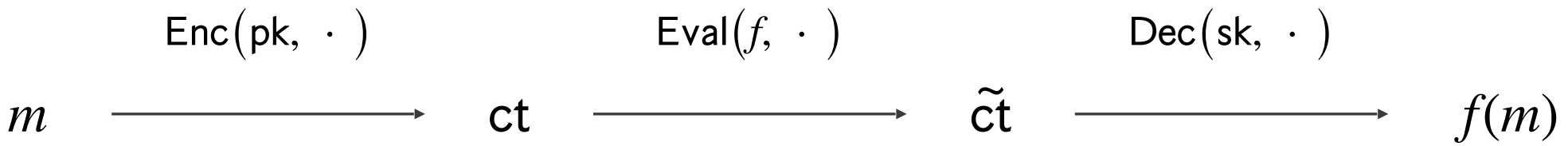
Fully Homomorphic Encryption from Subset Sum

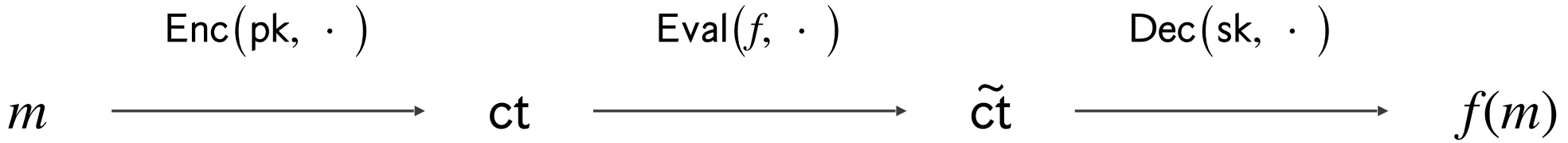
Giulio Malavolta Bocconi University
Zayd Maradni MPI-SP

Fully-homomorphic encryption

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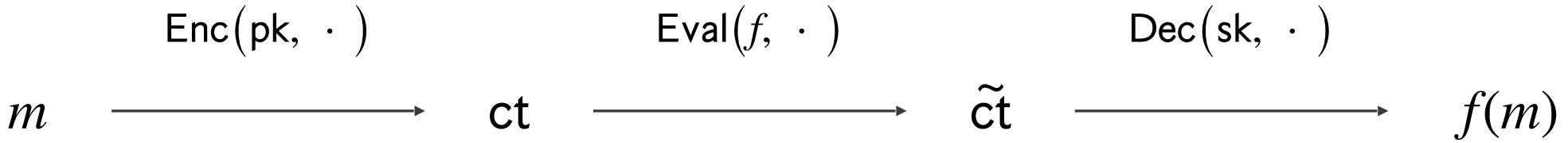


Fully-homomorphic encryption



Known constructions rely on **lattices (LWE)** [Gen09, BV11, BGV12, GSW13] or on **lattice isomorphism** [BMZ25] or on **indistinguishability obfuscation** [CLTV15].

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Can we build FHE from a different assumption?

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1 **FHE from subset sum**

If subset sum with density $\delta = \Theta\left(1/\log n\right)$ is hard, there is a leveled homomorphic encryption scheme for all bounded-depth circuits. Assuming circular security, it is fully homomorphic.

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$\mathsf{SS}(n, p^m)$: given $a_1, \ldots, a_n \in \mathbb{Z}_{p^m}$ and $t = \sum_{i=1}^{n} s_i a_i \bmod p^m$ for $\mathbf{s} \sim \{0,1\}^n$, find $\mathbf{s}$.

Distinguishing $(a_1, \ldots, a_n, t)$ from uniform is as hard as finding $\mathbf{s}$ [IN96]. **Density** $\delta := n/(m \log p)$.

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Column i of $\mathbf{A} \in \mathbb{Z}_p^{m \times n}$ is a_i in base p, and $\mathbf{A} \odot \mathbf{s} \in \mathbb{Z}_p^m$ is $\sum_i s_i a_i \bmod p^m$ in base p:

$$\mathbf{A} \odot \mathbf{s} = \mathbf{A}\mathbf{s} + e(\mathbf{A}, \mathbf{s}), \qquad \| e(\mathbf{A}, \mathbf{s}) \|_\infty \leq \sqrt{n} \log n$$

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Relation to lattices

$(\mathbf{s}, 1)$ is the shortest vector of the lattice

$$\mathscr{L} = \{\mathbf{x} \in \mathbb{Z}^{n+1} : [\mathbf{a} \quad -t] \cdot \mathbf{x} = 0 \bmod p^m\}$$

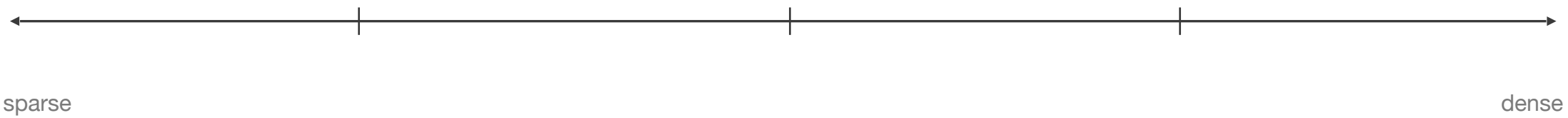
$\delta < 0.9408$: almost all instances reduce to uSVP [LO85, CJL+92]

For $\delta \approx 1/\log n$ this is uSVP_n, whereas LWE-based constructions use $\mathsf{uSVP}_{n^{1.5}}$.

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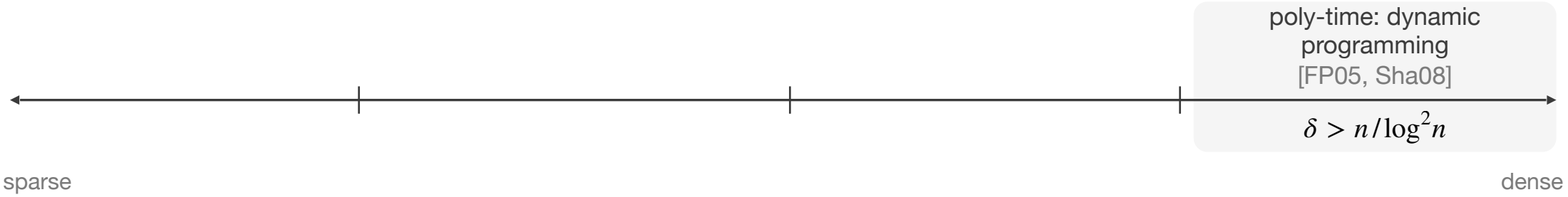
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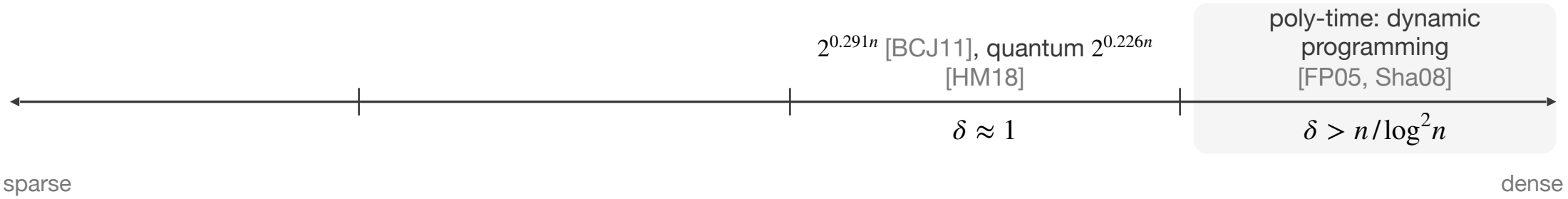
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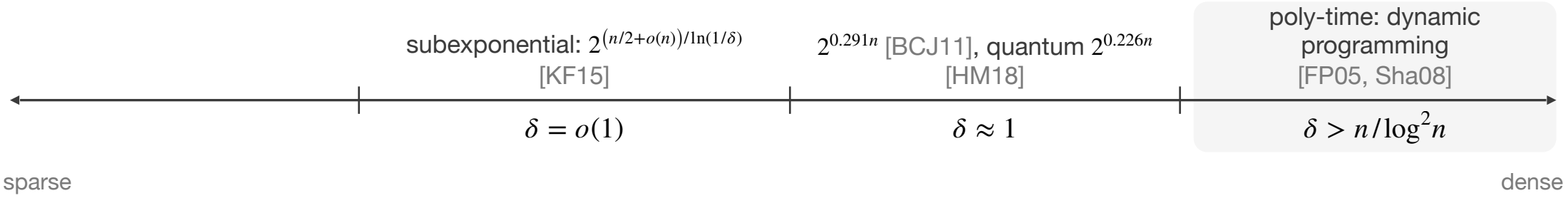
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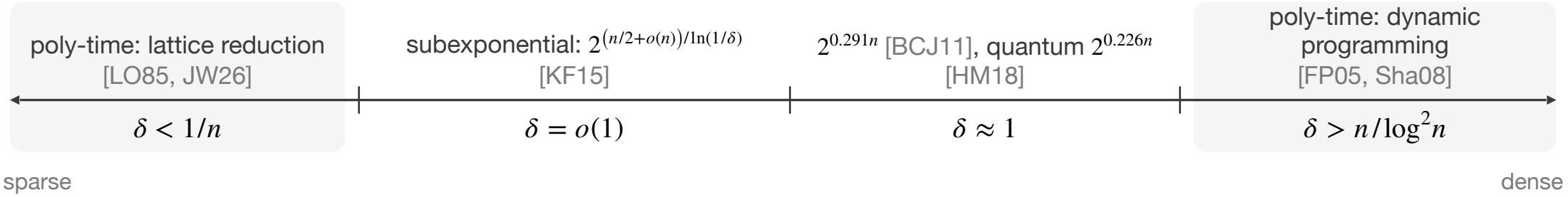
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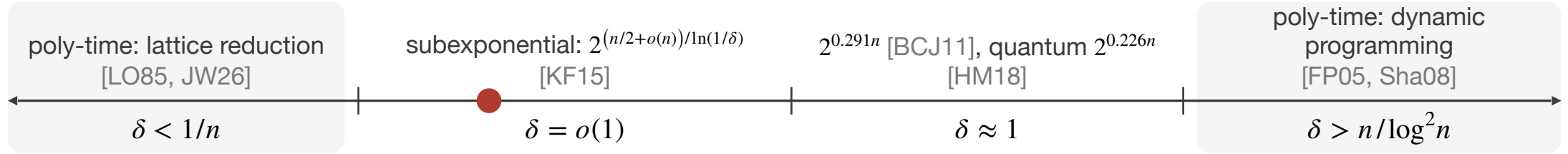
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sparse

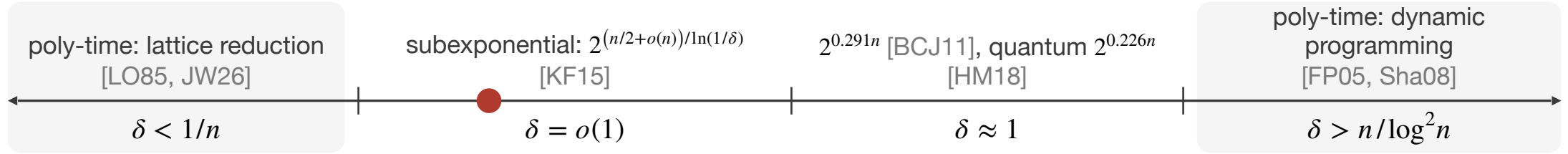
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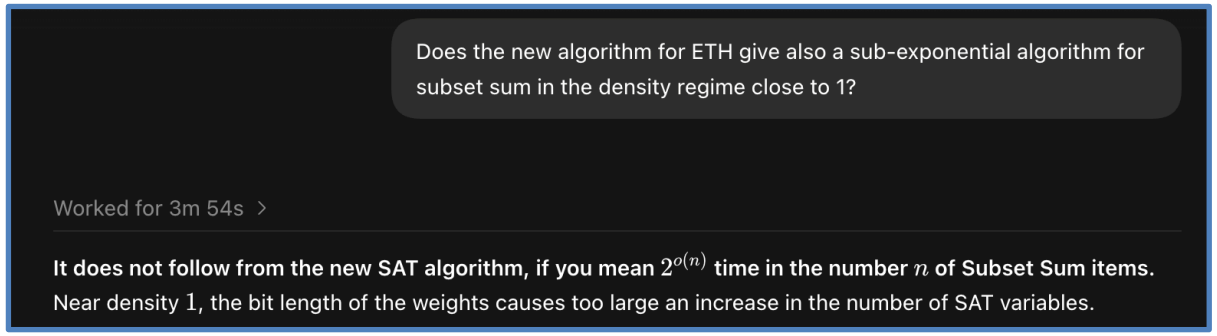
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KeyGen$\left(1^n\right)$: $\mathbf{A}' \sim \mathbb{Z}_p^{n \times n}$, $\mathbf{s} \sim \{0,1\}^n$

$$\mathsf{pk} := \mathbf{A} = \begin{bmatrix} \mathbf{A}' \\ \mathbf{s}^t \odot \mathbf{A}' \end{bmatrix} \in \mathbb{Z}_p^{(n+1) \times n}, \qquad \mathsf{sk} := [-\mathbf{s}^t \quad 1]$$

Enc$\left(\mathsf{pk},\, m \in \{0,1\}\right)$: $\mathbf{R} \sim \{0,1\}^{n \times (n+1)\log p}$

$$\mathbf{C} := \mathbf{A} \odot \mathbf{R} + m \cdot \mathbf{G}$$

with the gadget matrix $\mathbf{G} := \mathbf{I}_{n+1} \otimes \begin{bmatrix} 1 & 2 & \cdots & 2^{\log p - 1} \end{bmatrix}$

Dec$\left(\mathsf{sk}, \mathbf{C}\right)$: output

$$\left\lfloor \mathsf{sk} \cdot \mathbf{C} \cdot \mathbf{G}^{-1}\left(\begin{bmatrix} 0^n \\ p/2 \end{bmatrix} \right) \right\rceil_2$$

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Correctness

$\mathsf{sk} \cdot \mathbf{A} = \mathbf{s}^t \odot \mathbf{A}' - \mathbf{s}^t\mathbf{A}'$, so $\mathsf{sk} \cdot \mathbf{C} = m \cdot \mathsf{sk} \cdot \mathbf{G} + \mathbf{e}$ an approximate eigenvector, as in [GSW13]

Security

$\mathsf{SS}\left(n, p^{n+1}\right)$ hard ⇒ pk pseudorandom ⇒ $\mathbf{A} \odot \mathbf{R}$ pseudorandom ⇒ $\mathbf{C}$ pseudorandom.

Parameters

$p = \mathrm{poly}(n)$, hence $\delta = n/\left((n+1)\log p\right) = \Theta\left(1/\log n\right)$.

Homomorphic operations

$$\mathbf{C}_{\text{Add}} := \mathbf{C}_1 + \mathbf{C}_2$$

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Can we bootstrap?

Evaluating Dec gate by gate, the noise grows like $\left((n+1)\log p\right)^{\text{depth}}$: correctness forces a modulus p so large that the subset-sum instance lands in an weak, sparser density regime.

Bootstrapping

[BV14]: evaluate the decryption circuit as a branching program

Bootstrapping

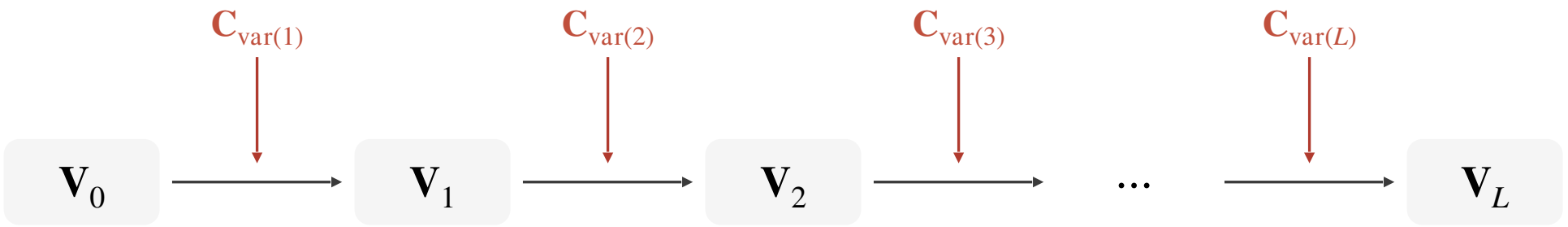
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Dec has depth $d = O\Big(\log\big((n+1)\log p\big)\Big)$. By Barrington's theorem it becomes a width-5 permutation branching program of length $L = 4^d = \mathrm{poly}(n)$, with state $\mathbf{V}_t = \big(\mathbf{V}_{t,1}, \ldots, \mathbf{V}_{t,5}\big)$ of encrypted bits.

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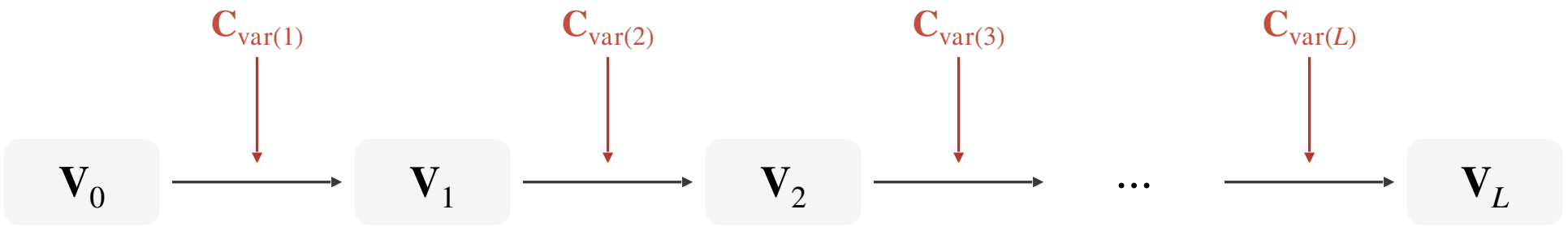
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Step t reads the input bit $x_{\mathrm{var}(t)}$ through its fresh encryption $\mathbf{C}_{\mathrm{var}(t)}$, with $\bar{\mathbf{C}} := \mathbf{G} - \mathbf{C}$ encrypting the complement:

$$\mathbf{V}_{t,i} = \bar{\mathbf{C}}_{\mathrm{var}(t)}\,\mathbf{G}^{-1}\Big(\mathbf{V}_{t-1,\gamma_0(t,i)}\Big) + \mathbf{C}_{\mathrm{var}(t)}\,\mathbf{G}^{-1}\Big(\mathbf{V}_{t-1,\gamma_1(t,i)}\Big)$$

The left ciphertext is always fresh, so the noise grows **additively**, by $4(n+1)^3\log^2 n \log^2 p$ per step: after $L = \mathrm{poly}(n)$ steps it is still $\mathrm{poly}(n) < p/4$. A polynomial p suffices, so the density stays $\Theta\big(1/\log n\big)$.

Thank you!

Questions?

